Lyna Hocine

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Education

Sorbonne University Sep 2025 – Sep 2027 (expected)
Master's in Algorithms, Artificial Intelligence, Interactions and Decision (AI2D) *Paris, France*

- **M1 (2025–26):** ranked 10/43 overall, 4/33 in semester 2; **M2:** 2026–27
- **Relevant Coursework:** AI for Adaptation of Multimodal Environments, Human-AI Interaction, AI for Robotics, Reinforcement Learning & Robotics, Multi-Agent Coordination & Consensus, Foundations of Multi-Agent Systems, Statistical & Probabilistic Models

University of Milan – 4EU+ Erasmus+ BIP Jun 2026
From Data to Systems: Designing and Deploying AI in Real-World Contexts *Milan, Italy*

- Intensive training on designing and deploying **AI systems in real-world contexts**, with a focus on challenges arising from real-world data.

Johannes Kepler University Feb 2025 – Jun 2025
Exchange Semester – Computer Science *Linz, Austria*

- **Relevant Coursework:** Unsupervised Machine Learning, SAT Solving, Engineering of AI-Intensive Systems

Sorbonne University Sep 2022 – Jun 2025
Bachelor's Degree in Computer Science *Paris, France*

Research Experience

ISIR – Sorbonne University 2026
Research Project – HARMONI + MCP: Agentic LLM Orchestration for HRI *Paris, France*

- Exposed **HARMONI**, a multimodal personalization framework for multi-user human-robot interaction, as structured tools through the **Model Context Protocol (MCP)**, allowing LLM agents to call its user-profile and context-retrieval functions during **multi-step interactions**.
- Connected LLM reasoning to HARMONI's **multimodal robot context and per-user memory** to enable context-aware interactions for **Enchanted Tools' Mirokaï** social robot.
- Implemented the MCP middleware in **Python** using FastMCP, LangChain, and Ollama for local LLM-based tool use.

ISIR – Sorbonne University / CNRS Jun 2026 – Jul 2026
Research Intern – Gesture-Based Human-Swarm Interaction *Paris, France*

- Studied **intuitive hand-gesture control** of robot collectives by extending a **C++ simulator**; implemented and evaluated decentralized behaviors (heading alignment, target tracking, formation, leader-follower) as the controllable targets.
- Analyzed the limits of **interpreting directional gestures** from noisy, incomplete local proximity sensing.

Sorbonne University 2026
Research Project – LLM-Driven Reward Function Generation for RL *Paris, France*

- Studied the **EUREKA** framework, an **iterative LLM loop** combining code generation, execution feedback, reflection, and evolutionary search to produce reward functions.
- Analyzed prompt synthesis, **automated feedback mechanisms**, and the interaction between LLM-generated rewards and policy optimization.
- Assessed whether LLMs can reduce the manual reward-engineering bottleneck in robot learning.

Technical Skills

Programming: Python, C++, C, Java, OCaml, SQL
ML / AI: PyTorch, NumPy, Pandas, scikit-learn
LLM Agents / Tooling: MCP, FastMCP, LangChain, Ollama, LLM tool use
Robotics / Simulation: human-robot interaction, gesture-based control, multi-agent simulation (C++)
Tools: Git, Linux, Bash
Languages: Arabic (Native), French (C1), English (C1), German (A1)

Research Interests

Human-Robot Interaction, Interactive Robot Learning, Multimodal and Human-Centered AI, Memory-Aware LLM Agents, Multi-Agent Systems